Advance Ambulance-Alert Corridor Simulator

Modeling Emergency Vehicle Corridor Dynamics & Advance Driver Awareness

Project Domain: Intelligent Transportation Systems (ITS)	**Date:** October 2026
Simulation Tool: Python 3 (NumPy, Matplotlib, Pytest)	**Corridor Setup:** 1D Discretized Highway (N=20)

Team Members & Roles

Member	Role	Key Responsibilities
Member A	Simulation Core	Mathematical dynamics, speed model, simulator core, test suite (simulator.py, test_simulator.py)
Member B	Experiments & Analytics	Parametric sweeps (H and α), CSV data logging, sensitivity plots (experiments.py)
Member C	Demo & Report	Interactive validation notebook, comprehensive technical report, presentation & viva prep

Academic Supervision & Institutional Details

Guide: [Faculty Guide Name] | **Department:** Computer Science & Engineering | **Institution:** College of Engineering

Problem Statement

In congested cities, ambulances get delayed because drivers react late to sirens. This project simulates a road corridor where drivers receive an advance alert that an ambulance is approaching. We compare ambulance travel time and stops with and without advance alerts to quantify the benefit of early awareness.

Objectives

- **To simulate ambulance movement** through a discrete traffic corridor under congestion.
- **To model the effect of advance alerts** on driver behavior and proactive lane clearing.
- **To quantify reduction in ambulance travel time** due to advance alerting.

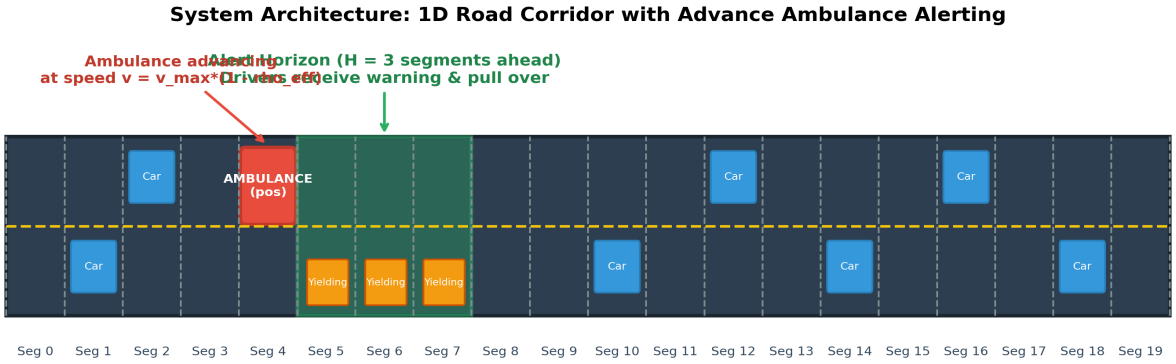
System Description & Simulation Methodology

System Description

- **Corridor:** 1D road divided into **N = 20 segments**.
- **Traffic Density:** Increases linearly from start to end (**0.2 to 0.8**), reflecting urban peak traffic bottlenecks.
- **Ambulance Speed Model:** Velocity decreases linearly as density rises: $v = v_{max} \cdot (1 - \rho)$, with $v_{max} = 1.0$ segment/step.
- **Alert Logic & Movement Mechanics:**
  – **Alert horizon H:** Segments ahead of ambulance that receive advance notification (default $H = 3$).
  – **Cooperation factor α:** Alerted drivers reduce effective density by $(1 - \alpha)$ (default $\alpha = 0.5$).
  – **Next-Segment Coupling:** Ambulance forward velocity is determined by the cleared effective density of the upcoming segment: $speed = ambulance_speed(\rho_{eff}[next_idx])$.

Corridor Architecture & Flow Model

"Road segments → Ambulance pos → Alerted segments ahead (clearing buffer zone)"



Simulation Approach

Ambulance movement is evaluated iteratively over discrete time steps until reaching the terminal corridor segment (idx = N-1) or safety limit $T_{max} = 200$. Two main scenarios are evaluated:

1. **No advance alert (Baseline):** Drivers only react at proximity; density remains at base levels $\rho_{base}[next_idx]$.
2. **With advance alert:** Drivers H segments ahead yield proactively, reducing effective density ρ_{eff} via cooperation factor α.

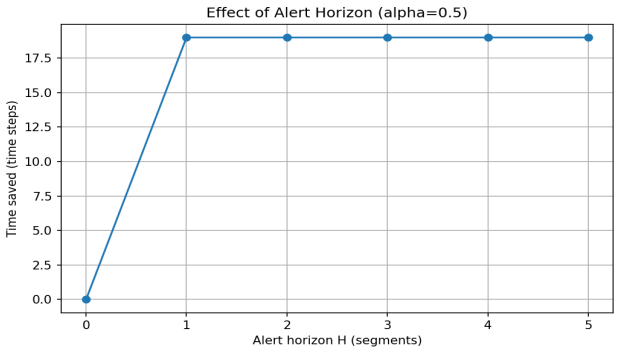
- **Metrics:** Travel time (time steps) and time saved $(\Delta t = time_no - time_alert)$.

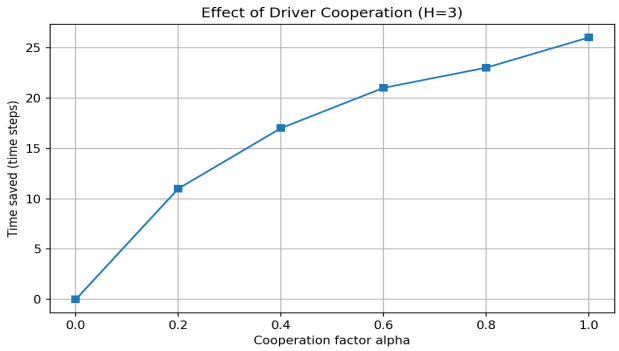
Results & Comparative Evaluation

Ambulance Movement: No Alert vs Advance Alert



Parametric Sweeps (Alert Horizon H & Driver Cooperation α)





Quantitative Summary Table (from results/summary.txt)

Scenario / Parameter	Travel Time	Time Saved	Measured Gain
Baseline (No Alert)	45 time steps	0 time steps	Baseline reference
Default Alert (H=3, α=0.5)	26 time steps	19 time steps	**42.22% faster** transit
Exp 1: H = 0 to 5 (α=0.5)	45 → 26 steps	0 → 19 steps	H≥1 unlocks full path clearing
Exp 2: α = 0.0 to 1.0 (H=3)	45 → 19 steps	0 → 26 steps	24.4% to 57.8% speedup

Observations:

- **Advance alerts reduce ambulance travel time by 42.22%** for typical parameters (dropping transit time from 45 down to 26 time steps).
- **Providing advance notice (H ≥ 1) completely mitigates upcoming segment bottlenecks**, cutting transit time from 45 to 26 steps.
- **Higher driver cooperation (α) monotonically and significantly increases benefit** (scaling from 11 time steps saved at α=0.2 up to 26 steps saved at α=1.0).

Real-World Mapping, Limitations & Future Work

Real-World Mapping

Advance alerts could be sent via:

- **Navigation apps (Google Maps, Waze, etc.):** Real-time in-app acoustic & visual pop-up notifications informing upstream vehicles to clear emergency lanes.
- **Digital road signboards:** Variable message signage (VMS) on overhead expressway gantries alerting drivers before intersection choke points.
- **SMS / Cell-broadcast to phones in a geo-area:** Localized sub-second mobile broadcasts targeting vehicles within the active corridor trajectory.

Limitations

- **Synthetic traffic pattern, not real data:** The corridor uses a synthetic linear density ramp rather than empirical detector data.
- **1D corridor, no lane-level detail:** Lateral lane changes, bottleneck merges, and intersection traffic signals are not explicitly resolved.
- **Driver behavior modeled simply:** Compliance is treated deterministically without driver reaction lag, vehicle physics, or distracted driver non-compliance.

Future Work

- **Use real city maps and traffic data:** Interface with microscopic simulators like SUMO or CORSIM using real GIS coordinates and sensor streams.
- **Model multiple ambulances and corridors:** Extend to urban network grids handling simultaneous emergency trips and route conflict resolution.
- **Integrate with real alert channels:** Pilot live API integrations with navigation apps and municipal smart traffic light controllers (Green Corridor).

Conclusion: The simulation demonstrates that proactive digital alert dissemination provides substantial travel time reductions (**42.22% measured improvement**) for emergency services in congested corridors. By ensuring early warning and driver cooperation, municipal ITS infrastructure can dramatically accelerate emergency response and safeguard the critical golden hour.